DO NOW, AO1 REVISION

Fleming's left-hand rule and motors

Magnetism and electromagnetism
Paper 2, Combined and Separate
13 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 7 starred (★) questions first.

	QUESTION	ANSWER	✓
1 ★	**Describe the motor effect.**	A current-carrying conductor in a magnetic field experiences a force.	☐
2 ★	**In Fleming's left-hand rule, state what the thumb represents.**	The direction of the force	☐
3 ★	**In Fleming's left-hand rule, state what the first finger represents.**	The direction of the magnetic field	☐
4 ★	**In Fleming's left-hand rule, state what the second finger represents.**	The direction of the current	☐
5 ★	**Give three ways to increase the force on a current-carrying wire in a magnetic field.**	Increase the magnetic flux density, the current, or the length of wire in the field.	☐
6	**Define magnetic flux density.**	A measure of the strength of a magnetic field.	☐
7	**State the standard unit of magnetic flux density.**	Tesla (T)	☐
8 ★	**Describe what happens to a current-carrying coil in a magnetic field.**	It rotates, because the forces on its two sides act in opposite directions.	☐
9	**Name the effect used in electric motors.**	The motor effect	☐
10	**State the equation for the force on a current-carrying conductor at right angles to a magnetic field.**	force = magnetic flux density × current × length (F = B I l)	☐
11	**Give two ways to reverse the force on a current-carrying wire in a magnetic field.**	Reverse the current or reverse the magnetic field.	☐
12	**Give two ways to make an electric motor spin faster.**	Increase the current, or use a stronger magnetic field.	☐

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	QUESTION	ANSWER	✓
13 ★	**Explain why a current-carrying wire in a magnetic field experiences a force.**	The magnetic field around the wire interacts with the field it is placed in.	☐